

MD RAKIBUL ISLAM RAIHAN

AI Systems Engineer | LLM Inference | GPU Optimization | Multimodal AI

2027 New Grad · Available July 2027 · Open to Relocation · Xi'an, China

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Dear Sir or Madam,

I am writing to apply for an AI Engineer position focused on LLM inference and multimodal research (2027 New Grad). I am an M.Eng. Software Engineering candidate at Northwestern Polytechnical University (Top 1%, 985/211, Chinese Government Scholarship), graduating July 2027. My work sits at the intersection of efficient multimodal generation and the systems that make such models deployable, and I am seeking a role - in an industrial research group or as a research engineer - where publication-quality work and production systems depth are both valued.

Research. I am first author of an IEEE TNNLS (Q1) paper under review: NVFP4-DiT, a 4-bit audio-guided video diffusion transformer achieving 4x memory reduction while maintaining generation quality. The work combines low-precision quantization (FP8/FP4/NVFP4) with multimodal cross-attention, and I benchmarked it on A800/H100 GPUs using PyTorch Profiler and Nsight Systems. My M.Eng. thesis focuses on multimodal deepfake detection using visual, audio, and temporal consistency analysis, and my B.Eng. thesis designed a distributed confidential-query framework using Apache Spark and CKKS homomorphic encryption - achieving 95% privacy preservation and a 20% performance improvement.

Research Interests. Efficient multimodal generation and understanding; low-precision (FP4/FP8/NVFP4) training and inference; and co-design of models with the serving stack - KV-cache management, PD disaggregation, and RDMA transport. I am drawn to problems where a modeling decision and a systems constraint must be solved together, and I would be glad to bring both my first-author track record and my production experience to a research group.

Systems. During my internship at InfIX.ai, I built production LLM serving on multi-node A800 and H800 GPU clusters using SGLang. I designed a 3-tier KV-cache offload pipeline (GPU HBM -> CPU DRAM -> SSD) over Mooncake RDMA, achieving a 92% cache hit rate and an 8.3x TTFT reduction (55s -> 6.7s) for long-context models. I patched SGLang source directly to fix production cache failures and deploy hybrid-attention Qwen models. I also built PD-disaggregated (prefill/decode) serving with RDMA KV sharing, raising stable concurrency from 192 to 256 requests, and profiled CUDA/Triton inference with Nsight Systems to localize GEMM bottlenecks and apply FP8 quantization for throughput gains. Beyond that, I diagnosed and resolved 12+ production incidents across RDMA, etcd, networking, storage, and distributed inference.

Open Source. I maintain ai-infra-portfolio, a curated suite of 10 production-grade, Dockerized AI-infra projects: LLM serving, observability, distributed training, MLOps, and API gateways; and train-launcher, a CLI that launches and supervises multi-node PyTorch training with fault tolerance and automatic retry.

Stack. Python (Proficient) | C/C++ | Bash | SQL | SGLang | KV-cache & PD-disaggregation | Mooncake RDMA | FP8/FP4 quantization | CUDA | Triton | multi-GPU serving (4x A800) | PyTorch | Apache Spark | CKKS | Linux | Docker | Git | CI/CD.

What makes my profile unusual is the combination of algorithmic research and production systems depth - I understand both the modeling side and the hardware and serving constraints that make efficient multimodal AI necessary and hard. This lets me move fluidly between research experiments and deployed infrastructure. I would welcome the chance to discuss how my experience can contribute to your team or research group; my CV, NVFP4-DiT manuscript, and thesis are available on request. Thank you for your consideration.

Sincerely,

MD RAKIBUL ISLAM RAIHAN

M.Eng. Software Engineering, NWPU (Expected 2027)

MD RAKIBUL ISLAM RAIHAN

AI 系统工程师 | 大模型推理 | GPU 优化 | 多模态 AI

2027 届应届 · 2027 年 7 月到岗 · 可接受异地 · 中国·西安

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作品集: theraihanrakibb.github.io/Online-Portfolio



尊敬的先生/女士:

您好! 我写信申请 AI 工程师岗位(大模型推理与多模态研究, 2027 届应届)。我是西北工业大学软件工程硕士在读生(985/211, 专业前 1%, 中国政府奖学金), 预计 2027 年 7 月毕业。我的工作处于高效多模态生成与使其可部署的系统之间的交叉地带, 希望加入一个同时重视论文质量与生产系统深度的团队。

科研方面。 我是 IEEE TNNLS (Q1) 论文第一作者(审稿中): NVFP4-DiT——4 比特音频引导视频扩散 Transformer, 实现 4 倍显存降低且保持生成质量。该工作融合低精度量化(FP8/FP4/NVFP4)与多模态跨模态注意力机制, 并在 A800/H100 GPU 上使用 PyTorch Profiler 与 Nsight Systems 进行了基准测试。我的硕士论文方向为多模态深度伪造检测, 基于视觉、音频与时序一致性分析; 本科论文设计了基于 Apache Spark 与 CKKS 同态加密的分布式机密查询框架, 实现 95% 隐私保护与 20% 性能提升。

研究兴趣。 高效多模态生成与理解; 低精度(FP4/FP8/NVFP4)训练与推理; 以及模型与服务栈的协同设计——KV-cache 管理、PD 分离与 RDMA 传输。我尤其关注建模决策与系统约束必须共同求解的问题, 并希望将我的第一作者成果与生产系统经验带入研究团队。

系统方面。 在 InfX.ai 实习期间, 我在 A800 与 H800 多节点集群上基于 SGLang 构建生产级大模型服务。我设计了三级 KV-cache 卸载方案(GPU HBM → CPU DRAM → SSD), 通过 Mooncake RDMA 实现 92% 缓存命中率与 8.3 倍 TTFT 降低(55s → 6.7s)。我直接修改了 SGLang 源码, 修复生产缓存故障并支持混合注意力 Qwen 模型部署。我还构建了 PD 分离(prefill/decode)推理架构, 基于 RDMA 共享 KV, 将稳定并发从 192 提升至 256; 并使用 Nsight Systems 定位 GEMM 瓶颈, 通过 FP8 量化提升吞吐。此外, 我排查并解决了 12+ 起生产故障(RDMA、etcd、网络、存储与分布式推理)。

开源贡献。 我维护 ai-infra-portfolio——收录 10 个生产级、Docker 化的 AI 基础设施项目, 涵盖大模型服务、可观测性、分布式训练、MLOps 与 API 网关; 以及 train-launcher (多节点 PyTorch 训练启动与容错重试 CLI)。

技术栈。 Python (熟练) | C/C++ | Bash | SQL | SGLang | KV-cache 与 PD 分离 | Mooncake RDMA | FP8/FP4 量化 | CUDA | Triton | 多卡服务 (4× A800) | PyTorch | Apache Spark | CKKS | Linux | Docker | Git | CI/CD。

我的独特优势在于算法研究与生产系统深度的结合——既理解建模层面, 也理解使高效多模态 AI 成为必要且困难的硬件与服务约束。这种双重视角使我能够在研究实验与已部署基础设施之间自由切换。期待有机会进一步沟通; 我的简历、NVFP4-DiT 论文与毕业论文均可应要求提供。感谢您的阅读与考虑!

此致 敬礼

MD RAKIBUL ISLAM RAIHAN

软件工程硕士, 西北工业大学 (预计 2027 年毕业)